

# Mesh-to-Mesh Transfer Errors in BDF2 Unsteady Flow Simulations

## Abstract

Discrete remeshing in a time-accurate CFD calculation changes both the spatial discretization and the histories supplied to a multistep integrator. This work separates the resulting errors through static transfer tests, repeated Euler remeshing, controlled BDF2 restarts, and fixed-body RANS calculations with SA and SST turbulence transport. Nearest-neighbor Taylor transfer, barycentric interpolation, and continuous Galerkin projection recover their nominal first- to third-order smooth-field behavior. Under repeated Riemann remeshing, the dominant stress-test error is cumulative profile broadening rather than feature translation: in the high-frequency setting, NN1 and BAR2 broaden the density transition by 21.4 and 3.24 cells at the baseline deformation and by 26.4 and 22.9 cells when the displacement bound is doubled, whereas GP2 changes the width by at most 0.06 cell; the contrast is reproduced by three independent deformation realizations. A bounded global correction reduces the NN1 density-integral defect to roundoff without removing this broadening, in all three realizations. In a controlled Euler NACA 0012 restart, perturbing only the older BDF2 history produces a finite load response even with an exact latest history, and the measured directional response ratios recover the BDF2 coefficients to within a few tenths of a percent in norm. The residual audit further shows that slip-wall enforcement modifies the wall-normal momentum component introduced by Galerkin transfer. In RANS, GP2 reduces the monitored turbulence-inventory defects by factors of roughly 37–247 relative to NN2, but no aerodynamic ranking is resolved: for SA the paired lift/drag difference is converged with respect to the dual-time iteration cap yet changes by more than its own magnitude when the physical step is halved and again when it is quartered; for SST the  $\omega$  residual floors at  $10^{-7}$  on the moved grids and the paired difference changes by 78% of itself between iteration caps. The results show that interpolation order, boundedness, integral preservation, multistep-history consistency, boundary processing, and receiving-solver convergence are distinct requirements; none alone predicts the aerodynamic consequence of remeshing.

## 1 Introduction

Unsteady Reynolds-averaged Navier–Stokes (RANS) calculations use mesh adaptation to follow evolving flow structures without maintaining fine resolution throughout the domain. When adaptation regenerates connectivity, the flow and turbulence histories must be transferred to the new discretization before time integration can continue. The resulting calculation combines spatial approximation, solution transfer, and temporal integration; improving one component does not necessarily reduce their combined error. Time-accurate anisotropic adaptation and space–time adaptation for implicit RANS calculations illustrate this coupling [1, 2].

Two questions arise at each mesh change. Does the transfer represent the source field accurately on the target mesh, and does the transferred state permit a consistent continuation of the target discrete evolution? Polynomial reproduction and integral conservation address the first question only partially. A transfer can preserve a total integral while redistributing the field in a way that changes a residual balance or an aerodynamic load. Conversely, an accurate pointwise interpolant

can introduce a small global imbalance. Repeated transfers may accumulate either effect, depending on the error’s spatial structure and its interaction with the evolving flow.

The distinction between interpolation accuracy and preservation of a discrete state is well established. Common-refinement Galerkin projection provides an integral-preserving, best- $L_2$  transfer between finite-element spaces [3, 4], while constrained variants are needed when additional discrete properties must be preserved [5]. Balance-preserving transfer in shallow-water models provides a related example: an accurate interpolant need not preserve a steady discrete balance [6, 7]. Here these ideas are examined in the specific setting of a vertex-centered finite-volume BDF2 restart and its subsequent aerodynamic response.

The comparison uses three established constructions: nearest-neighbor Taylor transfer, barycentric interpolation on a simplicial representation of the source mesh, and continuous Galerkin projection. Their nominal orders range from one to three. A bounded global correction is also applied to the pointwise methods, allowing the effect of enforcing a total integral to be assessed without replacing the underlying interpolation rule. The analysis distinguishes the consistent finite-element integral from the nodal integral actually used by the receiving solver, particularly when a quadratic projected field is reduced to vertex-only output.

The numerical argument proceeds from interpolation verification to repeated transport and then to the response of the receiving flow solver. A two-dimensional Euler Riemann problem isolates accumulation under changes in remeshing frequency and deformation amplitude. A controlled Euler NACA 0012 restart separates the contributions of the two BDF2 histories and examines boundary-condition processing and delayed aerodynamic response. The final application retains a fixed airfoil but adds SA and SST turbulence transport, viscous stresses, and repeated random mesh replacement. This sequence is designed to determine which transfer properties survive contact with a complete time-accurate CFD pipeline, rather than to establish a universal ranking of interpolation operators. Continuous mesh motion and moving bodies are outside the present scope.

## 2 Solution-transfer operators

### 2.1 Representation and reconstructed derivatives

The source and target meshes,  $A$  and  $B$ , cover the same discretized domain  $\Omega$ . Their nodal values are  $U_A$  and  $U_B$ , and  $M_M = \text{diag}(w_i^M) \otimes I$  contains the positive control-volume weights of the vertex-centered flow solver on mesh  $M$ . For one component  $\phi$ , the solver’s integral is

$$Q_M(\phi_M) = \sum_i w_i^M \phi_{M,i}. \quad (1)$$

All compared transfers return values at the solver vertices. Density, momentum, and total energy are transferred in conserved-variable form; nonlinear products are not reconstructed from separately interpolated primitive variables. Nodal sampling and control-volume averaging are different representation operators, even when the same weights are used to monitor an integral.

Nearest-neighbor corrections and the quadratic source representation require derivatives recovered from surrounding nodal data. A weighted least-squares fit determines the coefficients in

$$\phi_j - \phi_i \simeq g_i \cdot r_{ij} + \frac{1}{2} r_{ij}^T H_i r_{ij}, \quad r_{ij} = x_j - x_i. \quad (2)$$

The degree-one fit omits the Hessian; the degree-two fit retains it. With a full-rank, suitably scaled stencil and smooth data, polynomial reproduction gives an  $O(h)$  gradient error for the linear fit and  $O(h^2)$  gradient and  $O(h)$  Hessian errors for the quadratic fit. These are conditional estimates:

irregularity, anisotropy, rank deficiency, and weighting can degrade derivative recovery [8, 9, 10, 11]. The third-order variants therefore use degree-two reconstruction. Their accuracy is checked numerically rather than inferred from the requested polynomial degree alone.

## 2.2 Nearest-neighbor and barycentric interpolation

Nearest-neighbor transfer selects the closest source vertex  $n(j)$  to each target vertex and evaluates a Taylor expansion there. With  $r_j = x_j^B - x_{n(j)}^A$ , the three variants are

$$(\mathcal{I}_{AB}^{\text{NN1}} \phi_A)_j = \phi_{A,n(j)}, \quad (3)$$

$$(\mathcal{I}_{AB}^{\text{NN2}} \phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j, \quad (4)$$

$$(\mathcal{I}_{AB}^{\text{NN3}} \phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j + \frac{1}{2} r_j^T H_{n(j)} r_j. \quad (5)$$

Thus NN1 copies a value, NN2 accounts for local slope, and NN3 also accounts for curvature. No containing-element search is required. For  $|r_j| = O(h)$  and accurate derivatives, their smooth-field errors are respectively  $O(h)$ ,  $O(h^2)$ , and  $O(h^3)$ . The selected donor changes discontinuously across source-node Voronoi boundaries, so a smooth displacement need not produce a smooth change in the interpolation error.

Barycentric interpolation instead locates the source simplex containing the target point. A conforming triangulated or tetrahedralized view of the original mesh supplies this search geometry, using original vertices rather than an augmented control-volume point set. If the containing simplex has vertices  $x_{i_r}^A$ , its barycentric coordinates satisfy

$$x_j^B = \sum_{r=1}^{d+1} \lambda_r x_{i_r}^A, \quad \sum_r \lambda_r = 1, \quad \lambda_r \geq 0. \quad (6)$$

They express the target point as a geometric weighted average of the simplex vertices. Applying the same weights to the field gives

$$(\mathcal{I}_{AB}^{\text{BAR2}} \phi_A)_j = \sum_r \lambda_r \phi_{A,i_r}, \quad (7)$$

$$(\mathcal{I}_{AB}^{\text{BAR3}} \phi_A)_j = \sum_r \lambda_r \left[ \phi_{A,i_r} + \frac{1}{2} g_{i_r} \cdot (x_j^B - x_{i_r}^A) \right]. \quad (8)$$

BAR2 is affine-exact and already second order. For a quadratic field with exact vertex gradients, the half-gradient term in BAR3 cancels the quadratic interpolation remainder, yielding third-order consistency. The factor 1/2 follows from this cancellation rather than from calibration. Both expressions are continuous across a conforming simplex interface when shared vertices have unique values and gradients. Neither enforces Eq. (1) between meshes.

Limiting is separate from nominal order. A factor  $0 \leq \Phi_i \leq 1$  may scale both derivative terms in NN3, so  $\Phi_i = 0$  recovers NN1; in BAR3 it scales the gradient correction and recovers BAR2, not a piecewise-constant interpolant. The smooth verification uses unlimited reconstruction. The discontinuous tests compare unlimited and Barth–Jespersen-limited transfer [12]; differentiable and higher-order limiter constructions provide related alternatives [13, 14, 15]. Damping a quadratic with a factor chosen from linear increments alone does not establish boundedness at every evaluation point. The measured extrema and the final thermodynamic admissibility of the state remain necessary diagnostics.

## 2.3 Galerkin projection and common-refinement integration

Galerkin projection determines a coupled target field rather than evaluating each target point independently. Let  $V_B^p$  be the continuous degree- $p$  Lagrange space on the target simplicial mesh. The projection of a source representation  $\phi_A^h$  satisfies

$$(\phi_B^h - \phi_A^h, v_B)_{L_2(\Omega)} = 0 \quad \forall v_B \in V_B^p, \quad (9)$$

and minimizes the  $L_2$  distance to that source representation [3, 4]. For  $\phi_B^h = \sum_j v_j N_j^B$ , this gives

$$H_B v = b, \quad (H_B)_{ij} = \int_{\Omega} N_i^B N_j^B dx, \quad b_i = \int_{\Omega} N_i^B \phi_A^h dx. \quad (10)$$

The consistent mass matrix  $H_B$  is sparse and positive definite. Its off-diagonal entries couple basis functions sharing target elements; solving the system determines all coefficients simultaneously. It is distinct from the diagonal finite-volume matrix  $M_B$  used in time integration.

The load vector is the cross-mesh operation. On one target element,  $N_i^B$  has a fixed polynomial expression, but the source field may cross several source-element boundaries and change polynomial expression within that element. Consequently,

$$b_i = \sum_{K_B} \sum_{K_A: |K_A \cap K_B| > 0} \int_{K_A \cap K_B} N_i^B \phi_A^h dx. \quad (11)$$

Each intersection fixes both polynomial expressions and permits exact polynomial quadrature. These intersections form the source–target common refinement; they may be integrated locally without storing a complete additional mesh. Coverage must be complete and nonoverlapping. Products require degree-two quadrature for  $P^1/P^1$  transfer and degree-four quadrature for  $P^2/P^2$  transfer on affine simplices [16, 17]. This geometric integration and the mass-matrix solve distinguish projection from the point searches in NN and BAR.

GP2 uses  $P^1$  source and target spaces. GP3 uses  $P^2$  spaces, introducing edge coefficients internally. With  $d_{ij} = x_i - x_j$  and  $\bar{H}_{ij} = (H_i + H_j)/2$ , the source midpoint value is reconstructed as

$$\phi_{ij} = \frac{1}{2}(\phi_i + \phi_j) - \frac{1}{8}d_{ij}^T \bar{H}_{ij} d_{ij}. \quad (12)$$

For a quadratic, the subtracted curvature term is exactly the difference between the endpoint average and the midpoint value. GP3 then solves for the complete quadratic target field, but only its original-vertex values are supplied to the flow solver. Nominal  $L_2$  approximation orders are two and three under the usual smoothness and mesh-stability assumptions; nodal accuracy after this complete pipeline is assessed separately.

On mixed-element grids, BAR and GP use conforming subdivisions based on a deterministic shared-face diagonal rule. The approximation is  $P^p$  on the simplices, not  $Q^p$  on the native quadrilaterals or hexahedra. A valid conforming split and the chosen control-volume weights must therefore be retained consistently in geometric and integral checks.

## 2.4 Which integral is conserved?

For an unconstrained target space containing constants, Eq. (9) preserves the integral of the complete finite-element field. Coordinate functions also belong to continuous  $P^1$ , and quadratic coordinates belong to  $P^2$ . Exact projection therefore preserves the corresponding consistent moments before constraints, sampling, or limiting. This is not finite-volume conservation over arbitrary target cells: their indicator functions generally do not belong to the continuous test space. Nor is the inverse

mass matrix a local transfer operator. Equality of source and target geometry on one subregion does not by itself make the projection the identity there.

For GP2, the consistent and solver-lumped integrals coincide when  $w_i^M = \int N_i^M dx$  on both meshes. For GP3, source lifting and target vertex restriction each introduce a representation-dependent difference. In particular, a third-order projected field does not imply an  $O(h^3)$  lumped-integral defect. A simple quadratic counterexample is given in Appendix A.1: exact  $P^2$  projection can still produce an  $O(h^2)$  error after restriction to vertex values and  $P^1$  lumped integration. The consistent projection, the stored restart field, and any repaired field are consequently distinguished throughout the comparison.

A global correction can enforce a chosen solver integral independently of the transfer construction. For a raw nodal field  $\phi_B^r$ , it seeks a correction  $c$  such that

$$\sum_j w_j^B c_j = Q_A(\phi_A) - Q_B(\phi_B^r), \quad \ell_j \leq \phi_{B,j}^r + c_j \leq u_j. \quad (13)$$

The implemented repair redistributes the defect over nodes with available capacity, freezing saturated nodes [18, 19]. Feasibility requires the requested integral to lie between the weighted lower and upper bounds, with fixed boundary values included. Source-range bounds on individual conserved components do not alone guarantee positive pressure. Accuracy is retained only if the correction is sufficiently small in the accuracy norm; a small total defect does not itself bound every nodal increment.

The optional NN/BAR correction is denoted by the suffix GC. GP additionally uses a projection-level limiter: source-vertex values in overlapping elements define  $[\ell_j, u_j]$ , widened by  $\tau$  times the global source range. Clipping is followed by redistribution preserving a selected lumped total. Hard limiting sets  $\tau = 0$ ; soft limiting allows a controlled excursion. Once active, this nonlinear post-processing need not preserve Galerkin orthogonality, first moments, or the best-approximation property. For GP2, preserving its lumped total preserves its consistent mass only under the weight compatibility above. For vertex-only GP3, no such implication holds automatically. These qualifications matter when interpreting the method labels in the numerical results.

### 3 BDF2 response to a mesh change

#### 3.1 Reference trajectories and history mismatch

Between remeshing events, the mesh is fixed and the semi-discrete equations are

$$M_M \dot{U}_M + F_M(U_M, t) = 0. \quad (14)$$

The residual can contain Euler, viscous, or RANS terms; the state must include every transported variable required by the time integrator. The constant-step BDF2 equation on the target mesh is

$$\Phi_B(Y; V, W) = \frac{M_B}{\Delta t} \left( \frac{3}{2}Y - 2V + \frac{1}{2}W \right) + F_B(Y, t^{n+1}) = 0. \quad (15)$$

This form is not a general moving-volume ALE formula. Continuously moving meshes require consistent geometric history terms and a discrete geometric conservation law [20, 21]; those are outside the fixed-mesh-between-transfers derivation.

Let  $U_A^k$  and  $U_B^k$  be native trajectories for the same physical problem, computed on the two meshes. The transferred histories and their target-referenced discrepancies are

$$\hat{U}_B^k = \mathcal{I}_{AB}(U_A^k), \quad \varepsilon^k = \hat{U}_B^k - U_B^k, \quad k = n, n-1. \quad (16)$$

Here  $\mathcal{I}_{AB}$  includes all limiting, correction, and restart-field processing. Let  $\Pi_M u^k$  represent a common exact field in the solver's chosen data representation, and  $e_M^k = U_M^k - \Pi_M u^k$ . An exact decomposition, valid also for nonlinear transfer, is

$$\begin{aligned} \varepsilon^k = & \underbrace{\mathcal{I}_{AB}(\Pi_A u^k) - \Pi_B u^k}_{\text{transfer consistency}} \\ & + \underbrace{\mathcal{I}_{AB}(U_A^k) - \mathcal{I}_{AB}(\Pi_A u^k) - e_B^k}_{\text{propagated source error and target error}}. \end{aligned} \quad (17)$$

Only for a linear transfer may the second line be replaced by  $\mathcal{I}_{AB}e_A^k - e_B^k$ .

Thus a native-target reference does not make  $\varepsilon^k$  a pure interpolation error. It puts both states into the same target evolution problem while retaining their pre-existing discretization mismatch. For an output  $G$ , the decomposition

$$\begin{aligned} G_B(\hat{U}_B^k) - G_A(U_A^k) = & [G_B(U_B^k) - G_A(U_A^k)] \\ & + [G_B(\hat{U}_B^k) - G_B(U_B^k)] \end{aligned} \quad (18)$$

separates native mesh sensitivity from the target-referenced restart discrepancy, not from an independently known consistency error. Comparisons of different transfers from the same source histories to the same target provide the controlled method comparison.

### 3.2 Exact history forcing and local state response

At any common trial state  $Y$ , changing only the histories gives the exact identity

$$\Phi_B(Y; U_B^n + \varepsilon^n, U_B^{n-1} + \varepsilon^{n-1}) - \Phi_B(Y; U_B^n, U_B^{n-1}) = -\frac{M_B}{\Delta t} \eta^n, \quad \eta^n = 2\varepsilon^n - \frac{1}{2}\varepsilon^{n-1}. \quad (19)$$

This identity does not require a linear spatial residual or small transfer errors. It tests history assembly provided both residual evaluations use exactly the same trial state and residual convention.

For the solved states, define  $\delta^{n+1} = \hat{U}_B^{n+1} - U_B^{n+1}$ . Let  $r$  and  $\hat{r}$  be the remaining BDF2 algebraic residuals. Subtraction and expansion about the native target state give

$$A_B \delta^{n+1} = M_B \eta^n + \Delta t(\hat{r} - r) - \Delta t \mathcal{N}_B(\delta^{n+1}), \quad A_B = \frac{3}{2}M_B + \Delta t J_B, \quad (20)$$

where  $J_B = \partial F_B / \partial U$  and  $\mathcal{N}_B$  is the nonlinear Taylor remainder. With a locally Lipschitz Jacobian,  $\mathcal{N}_B(\delta) = O(\|\delta\|^2)$ . If  $A_B$  is locally invertible, the perturbations are small, and algebraic errors are negligible,

$$\delta^{n+1} = R_B \eta^n + O(\Delta t \|\eta^n\|^2), \quad R_B = A_B^{-1} M_B. \quad (21)$$

The constants are local to the mesh, state, time step, and norm; they are not assumed uniform over arbitrary anisotropic mesh families. Active limiter switches or insufficient inner convergence may invalidate this smooth-response approximation, but not Eq. (19).

The steady experiment uses one spatial discrepancy  $\varepsilon^* = \mathcal{I}_{AB}U_A^* - U_B^*$  and assigns it independently to the two histories:

| History configuration | $\varepsilon^n$ | $\varepsilon^{n-1}$ | $\eta^n$           |
|-----------------------|-----------------|---------------------|--------------------|
| Native                | 0               | 0                   | 0                  |
| Full                  | $\varepsilon^*$ | $\varepsilon^*$     | $3\varepsilon^*/2$ |
| Latest only           | $\varepsilon^*$ | 0                   | $2\varepsilon^*$   |
| Older only            | 0               | $\varepsilon^*$     | $-\varepsilon^*/2$ |

These diagnostic restarts predict

$$\delta_n \simeq \frac{4}{3}\delta_{\text{full}}, \quad \delta_{n-1} \simeq -\frac{1}{3}\delta_{\text{full}}, \quad \delta_{\text{full}} \simeq \delta_n + \delta_{n-1}. \quad (22)$$

The older-only case distinguishes the two-history forcing from a latest-state diagnostic. All three perturbations nevertheless use the same spatial direction: recovering their ratios is a test of directional response, not a complete validation of  $R_B$  for arbitrary history fields.

### 3.3 Aerodynamic response and propagation

For a differentiable aerodynamic functional with gradient  $g_B^{n+1}$ , the first-step discrepancy is

$$\Delta G^{n+1} = (g_B^{n+1})^T R_B \eta^n + O(\|\eta^n\|^2). \quad (23)$$

Equivalently,  $A_B^T z_B = g_B^{n+1}$  gives  $\Delta G^{n+1} = z_B^T M_B \eta^n$  to first order [22]. The error is weighted by the target dynamics and the output sensitivity, not by a constant conservation test. Conservation of the source integral constrains only

$$\mathbf{1}^T M_B \varepsilon^k = Q_A(U_A^k) - Q_B(U_B^k), \quad (24)$$

which need not vanish because the native integrals may differ.

A slope discrepancy must include the output difference already present at the transfer time:

$$\frac{\Delta G^{n+1} - \Delta G^n}{\Delta t} \simeq \frac{(g_B^{n+1})^T R_B \eta^n - (g_B^n)^T \varepsilon^n}{\Delta t}. \quad (25)$$

Dividing the first-step output difference alone by  $\Delta t$  is generally not this quantity. Cancellation between the two terms also prevents a universal force-kink scaling from being inferred from interpolation order alone.

For later steps, linearization along the native target trajectory gives

$$\begin{bmatrix} \delta^{k+1} \\ \delta^k \end{bmatrix} \simeq \begin{bmatrix} 2R_B^{k+1} & -R_B^{k+1}/2 \\ I & 0 \end{bmatrix} \begin{bmatrix} \delta^k \\ \delta^{k-1} \end{bmatrix}. \quad (26)$$

Repeated application connects the initial histories to a delayed output. This explains why a first-step force difference need not bound the later excursion. A dominant disturbance originating away from the airfoil may become measurable after acoustic or convective transport, but neither the inverse mass matrix of projection nor an implicit time step has compact support. Distance-dependent arrival is therefore measured by amplitude thresholds, not by a claim of strict first-nonzero causality.

## 4 Numerical experiments and assessment

### 4.1 Static transfer and repeated Euler transport

The static tests separate transfer error from flow-solver error. The asymmetric smooth density is

$$\phi = \rho_\infty \left[ 2 + 0.3e^{0.8x/L_x + 0.5y/L_y} \sin(2\pi x/L_x) \sin(2\pi y/L_y) \right]. \quad (27)$$

The field is transferred to structured quadrilateral and independently generated mixed triangular/quadrilateral target meshes at nominal resolutions 64, 128, and 256; the source mesh has half the target nominal resolution. Errors are relative nodal density  $L_2$  differences from the analytic value at

the target points. The effective spacing is defined as  $h_{\text{eff}} = N_{\text{nodes}}^{-1/2}$ ; on this fixed domain it is used only to define refinement ratios. The discontinuous test replaces the smooth field with a circular inclusion of radius  $0.25 \min(L_x, L_y)$  and density levels 1.2 and 3.6. Because the curved interface is not mesh-aligned, unlimited and limited transfers can be compared on a common unresolved discontinuity.

Consecutive and fitted orders are obtained from

$$p_{k,k+1} = \frac{\log(E_k/E_{k+1})}{\log(h_k/h_{k+1})}, \quad \log E \simeq p_{\text{fit}} \log h + b. \quad (28)$$

The three-level sequences are used to check consistency with the nominal orders, not to estimate mesh-independent asymptotic constants. A second smooth test on the structured Riemann mesh family keeps the mean displacement-to-spacing ratio near 0.057 and is used only as a consistency check on the implemented operators.

The unsteady Riemann calculation solves two-dimensional Euler flow on  $[0, 5] \times [0, 1]$ , with a uniform  $501 \times 101$ -node quadrilateral mesh,  $h = 0.01$ ,  $\Delta t = 10^{-3}$ , and 400 physical steps. At each event, the original mesh receives a new bounded random displacement in  $x$ , with boundary vertices fixed. Distortion is not accumulated geometrically from one event to the next. The bound is the minimum incident corner-triangle altitude divided by SafeFactor, denoted SF. Two transfers per block give 80 transfers for a ten-step interval or 400 for a two-step interval. SF4 is the baseline; SF2 doubles the displacement bound. A no-remeshing calculation supplies a fixed-mesh reference. This experiment measures the combined effect of changing spatial discretization and transferring the solution, not a pure interpolation error relative to an exact flow.

Interpolation-only integral changes are sampled immediately before and after each transfer and accumulated separately from the physical evolution. Density and energy are used because their reference integrals are away from zero. The stress-test dataset uses one common  $y$ -averaged profile extractor for every frequency and amplitude comparison: the transition position is obtained from the 50% crossing and its width from the 10–90% interval, both reported relative to the no-remeshing reference in units of the baseline cell size. The transition remains a broad density feature rather than an independently validated sharp shock. The frequency and correction matrix uses one deformation sequence; the NN1, BAR2, and GP2 entries at 400 transfers are repeated with two further random realizations at both amplitudes.

## 4.2 NACA 0012: independent meshes and controlled histories

The aerodynamic case is Euler flow over NACA 0012 at  $M_\infty = 0.1$  and  $\alpha = 6^\circ$ , with  $p_\infty = \rho_\infty = T_\infty = 1$ ,  $a_\infty = \sqrt{1.4} = 1.18322$ , and  $U_\infty = 0.118322$ . BDF2 dual-time integration uses  $\Delta t = 10^{-3}$ . Forces are pressure-only, with unit reference chord and area and moment reference  $x/c = 0.25$ . The nonzero steady computed drag is treated as a numerical load, not as skin friction or shock-induced wave drag; absolute coefficient changes are used.

The independent source and target meshes contain 16,255/32,077 and 22,508/44,583 nodes/triangles, respectively. Both share the same 370 airfoil vertices and 63-node farfield polygon at 100 chords. The surface geometry is therefore unchanged although the interior meshes differ. Each native steady solution is advanced with equal BDF2 histories to create its own reference trajectory. Transfers from  $A$  to  $B$  then populate the history configurations of Eq. (22). All comparisons of their solved responses use the native target trajectory at the same physical step.

State discrepancies use the nondimensional volume-weighted norm  $\|v\|_M = (v^T M v)^{1/2}$ , summed over the stored flow components. Because the farfield occupies most of the domain, this norm

emphasizes the global state and is complemented by near-wall and force diagnostics. Vector superposition is assessed by

$$e_{\text{sup}} = \frac{\|\delta_{\text{full}} - \delta_n - \delta_{n-1}\|_{M_B}}{\|\delta_{\text{full}}\|_{M_B}}, \quad (29)$$

with the analogous scalar expression for lift. Signed force ratios and positive state-norm ratios are kept distinct.

The solver exports  $-\Phi_B$ , including the temporal residual. Initial residuals are compared before the first update and only for cases sharing a trial state: older-only versus native at  $U_B^*$ , and latest-only versus full at  $\hat{U}_B^*$ . Their  $\Delta\eta$  values are respectively  $-\varepsilon^*/2$  and  $+\varepsilon^*/2$ . A dedicated double-precision residual probe is used for the final identity check. Because the NACA boundary is a slip wall, the residual is also decomposed into wall-normal and wall-tangential momentum components to distinguish the raw transferred history from the boundary-admissible state. First-step forces and subsequent 50-step summaries provide separate assessments of immediate and later response.

### 4.3 Repeated airfoil remeshing and localized disturbances

The repeated-remeshing experiment perturbs the source mesh between wall distances 0.01 and 20 chords while keeping the boundaries fixed. Thirty-nine mesh replacements occur, one every ten physical steps in the 400-step window. NN2 and GP2 are evaluated on the same two random deformation sequences, and force differences are formed relative to the no-remeshing trajectory. Least-squares slopes are quoted per 100 physical steps as descriptive summaries; the associated regression standard errors refer to each individual correlated time series and are not ensemble uncertainties.

A separate one-event experiment perturbs narrow wall-distance bands centered at  $d/c = 0.05$ , 0.10, and 0.20, then continues without further adaptation. Force onset is the first crossing of a selected fraction of that case’s peak response. Fixed pressure shells provide a less spatially integrated diagnostic through their root-mean-square pressure difference. A shell a distance  $d - r$  inside the launch band has the approximate acoustic lag  $(d - r)/(a_\infty \Delta t) \simeq 845(d - r)$ . This is a low-Mach scale estimate; actual characteristics depend on local velocity, sound speed, and propagation direction.

The no-remeshing trajectory is the reference for this localized-event experiment, whereas the mixed-history study uses an independently evolved target-mesh reference. Even though the meshes coincide outside the displaced band, the transfer is not guaranteed to remain strictly local. An immediate response is observed up to about  $0.08c$  inward of the band; those shells are excluded from the propagation fit. Projection coupling, limiter redistribution, the changed spatial operator, and the implicit update can all contribute to this near-instantaneous component.

### 4.4 Fixed-body RANS extension

The RANS extension uses the fixed NACA 0012 at  $M_\infty = 0.1$ ,  $\alpha = 6^\circ$ , and  $Re_c = 6 \times 10^6$  with Spalart–Allmaras (SA) and shear-stress-transport (SST) closures [23, 24]. The common hybrid mesh contains 23,271 vertices, 9,686 quadrilaterals in the wall layer and wake block, and 26,851 triangles, with 256 airfoil vertices and a farfield at  $100c$ . The first off-wall spacing is about  $5 \times 10^{-6}c$  and the quadrilateral region extends to approximately  $0.11c$ . The experiment is not a turbulence-model validation; it asks how repeated mesh replacement perturbs a nominally steady viscous solution when the turbulence histories are transferred with the mean flow.

The executed SU2 configurations use HLLC mean-flow fluxes with MUSCL reconstruction, scalar upwinding without MUSCL for the turbulence equations, CFL 6, FGMRES/ILU, and a maximum of 800 dual-time iterations per physical step. The convergence test begins after ten inner iterations

and targets  $\log_{10}(\text{rms } \rho) = -12$ . Runtime startup tables establish that the configured  $10^{-3}$  s step corresponds to the non-dimensional value  $\Delta t^* = 0.347224$  under `FREESTREAM_VEL_EQ_MACH`; with  $U_\infty^* = 0.1$ , the 400-step window spans 13.9 chord flow-through times. The first-order startup step is ten times smaller. Each closure has an 800-step no-remeshing reference initialized from the same restart histories.

At a repeated-remeshing event 12,602 vertices are movable and 10,669 remain fixed. No vertex with solver wall distance below  $0.01c$  is displaced; the motion mask is based on distance to the nearest airfoil node, so some vertices in the outer part of the quadrilateral layer do move. Boundary vertices are fixed. SF4 defines the baseline displacement and a minimum corner-area ratio of 0.2 is enforced; the mesh-quality audit shows valid grids throughout the event sequence. The displacement field depends only on the prescribed seed and event index, not on the evolving solution. For the seed-1 comparison, archived displacement hashes verify identical mesh sequences across SA/SST and NN2/GP2 cases.

The repeated comparison uses NN2 and limited GP2 with two deformation realizations and a replacement every ten physical steps. Density, momentum, total energy, and both BDF2 histories of the turbulence variables are transferred. SA transports  $\tilde{\nu}$ ; SST transfers density-weighted  $k$  and  $\omega$ , after which the receiving solver applies its primitive-variable positivity floors. Wall distance and eddy viscosity are recomputed. The monitored transfer integrals are therefore properties of the returned restart pipeline at the recorded stage, not conservation laws for subsequent turbulence evolution.

Total aerodynamic coefficients are compared with the matching no-remeshing reference. Pressure and friction contributions are additionally integrated from the exported surface fields using identical quadrature for case and reference. The repeated-remeshing force histories contain 398 distinct physical steps. Because the physical-step solve is part of the measured response, the number of inner iterations and terminal residuals are analyzed together with the transfer data rather than assumed negligible.

## 4.5 Evidence and reproducibility limits

The interpolation-order data and the Riemann and Euler NACA summary tables are versioned with the manuscript. The RANS archive additionally contains force histories, surface reductions, transfer-integral records, mesh-quality diagnostics, runtime normalization, code provenance, mesh-replay hashes, and inner-iteration histories. The repeated Riemann stress cases include three deformation realizations for NN1, BAR2, and GP2 at both deformation amplitudes. The RANS method comparison is accompanied by controls on the same prescribed remeshing sequence: dual-time iteration caps of 1600 and 6400, physical steps halved (SA and SST) and quartered (SA) with the events replayed at the same physical times, the same window on a finer viscous mesh, and no-transfer baselines on three meshes. These controls bound the conclusions below; the temporal sequence does not reach an asymptotic range, and the mesh sequence supplies one observed order per coefficient where the sequence is monotone.

# 5 Results

## 5.1 Interpolation order verification

Figure 1 shows the smooth-field refinement results. NN1 is first order; NN2, BAR2, and GP2 are consistent with second order; NN3 is consistent with third order. BAR3 and GP3 have fitted slopes above three over the three-level interval, approaching approximately 3.2 on the finest pair. This is

finite-range convergence behavior, not evidence of a higher designed order. On the finest mixed grid, the relative errors are approximately  $2.6 \times 10^{-7}$  for BAR3,  $3.4 \times 10^{-7}$  for GP3, and  $2.6 \times 10^{-6}$  for NN3. These differences resolve smooth interpolation performance on the tested mesh family; they do not predict a corresponding ranking of restarted flow solutions.

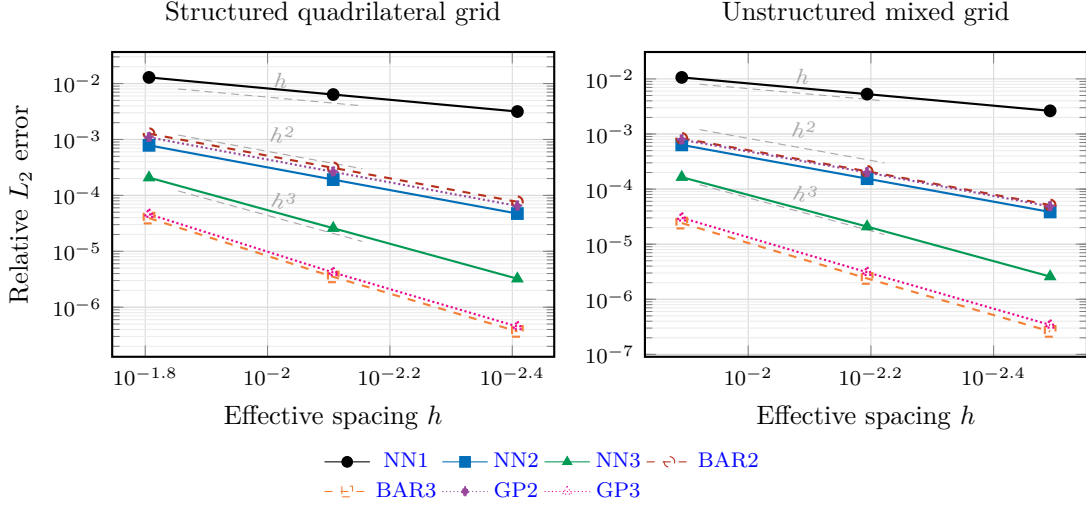


Figure 1: Convergence of the density-transfer error for the smooth asymmetric field. Curves show the relative nodal  $L_2$  error as the mesh is refined; gray guides correspond to  $O(h)$ ,  $O(h^2)$ , and  $O(h^3)$ .

## 5.2 Boundedness and integral preservation are independent constraints

The discontinuous transfer test separates three properties that coincide only in special cases: approximation error, boundedness, and preservation of the solver integral. Table 1 reports representative configurations on the structured  $64 \rightarrow 128$  transfer, the middle level of the refinement sequence. The second-order barycentric interpolant is bounded by construction on each containing simplex, whereas the unlimited derivative-based reconstructions generate new extrema. For NN2 the largest excursion is 0.803 above the upper plateau and 0.416 below the lower plateau; BAR3 reduces these excursions to 0.177 and 0.181, but does not remove them. Barth–Jespersen limiting removes the measured extrema. The same limiting step does not make NN2 conservative: where the limiter is active the reconstruction collapses to NN1, and the density-integral defect grows to the NN1 value of  $-4.46 \times 10^{-3}$  until the global correction is applied. Conversely, unlimited GP2 preserves its consistent integral to roundoff while remaining oscillatory. These observations rule out treating high order, boundedness, and conservation as interchangeable indicators of transfer quality.

The smooth single-transfer test reaches the same conclusion from the opposite direction. GC reduces the monitored density-integral defect to  $O(10^{-15})$  for every base transfer while changing the density  $L_1$  error by at most 0.8% at that resolution. It is therefore inexpensive in this smooth case, but its scalar constraint says nothing about where the correction is placed. The repeated-remeshing results below test that missing spatial information directly.

Table 1: Representative discontinuous-field transfers. Overshoot and undershoot are excursions beyond the exact density range  $[1.2, 3.6]$ ; the last column is the source-to-target defect in the solver’s lumped density integral. BJ denotes Barth–Jespersen limiting and GC the bounded global conservation correction.

| Transfer                         | Overshoot | Undershoot | Integral defect        |
|----------------------------------|-----------|------------|------------------------|
| NN2, unlimited                   | 0.8031    | 0.4157     | $-5.26 \times 10^{-5}$ |
| NN2, BJ                          | 0         | 0          | $-4.46 \times 10^{-3}$ |
| NN2, BJ+GC                       | 0         | 0          | $2.95 \times 10^{-14}$ |
| BAR3, unlimited                  | 0.1766    | 0.1811     | $-8.80 \times 10^{-7}$ |
| BAR3, BJ+GC                      | 0         | 0          | $5.59 \times 10^{-14}$ |
| GP2, hard limited                | 0         | 0          | $3.44 \times 10^{-15}$ |
| GP3, hard limited, vertex output | 0         | 0          | $-4.83 \times 10^{-7}$ |

Table 2: Effect of remeshing frequency, deformation amplitude, and global correction on the density-transition diagnostic. Width changes and absolute position errors are expressed in baseline cells ( $h = 0.01$ ). SF2 permits twice the displacement of SF4.

| Condition          | NN1 width | NN1+GC width | BAR2 width | GP2 width | max. $ \Delta x /h$ |
|--------------------|-----------|--------------|------------|-----------|---------------------|
| 80 transfers, SF4  | 0.57      | –            | 0.25       | 0.00      | 0.14                |
| 400 transfers, SF4 | 21.41     | 18.36        | 3.24       | 0.00      | 0.15                |
| 400 transfers, SF2 | 26.36     | 25.65        | 22.91      | 0.06      | 0.91                |

### 5.3 Repeated Euler remeshing: diffusion grows before position error

The Riemann diagnostics separate feature translation from broadening. At the baseline 80-transfer setting, all methods locate the density transition within 0.14 cell of the no-remeshing reference. The largest width change is 0.57 cell for NN1 and 0.25 cell for BAR2; the higher-order NN/BAR variants and both Galerkin variants change the width by at most 0.01 cell. The dominant failure mode under more frequent or larger mesh changes is therefore not a systematic translation of the feature but cumulative broadening.

Table 2 makes this distinction explicit. Increasing the number of transfers from 80 to 400 at SF4 expands the NN1 transition from 0.57 to 21.41 cells beyond the reference width, while its position error remains 0.10 cell. BAR2 broadens by 3.24 cells, whereas GP2 remains unchanged to the reported precision. Doubling the admissible deformation at 400 transfers increases the width changes to 26.36 cells for NN1 and 22.91 cells for BAR2; GP2 changes by only 0.06 cell. Even in this most aggressive comparison the reported position errors remain below one cell for all entries in the table. Two additional random deformation sequences reproduce this hierarchy (Table 6): at SF4 the width change is 21.4–22.9 cells for NN1, 3.1–3.2 cells for BAR2, and at most 0.01 cell for GP2; at SF2 the corresponding ranges are 25.0–27.1, 22.9–23.0, and 0.03–0.06 cells. Across all three realizations the feature-position error remains below 0.5 cell. The repeated result therefore concerns cumulative broadening, not a seed-specific displacement of the transition.

The global correction provides the cleanest separation between integral and spatial error. For NN1 at 400 transfers, it reduces the cumulative density-integral change from  $5.0 \times 10^{-4}$  to  $5.8 \times 10^{-14}$  at SF4 and from  $1.3 \times 10^{-3}$  to  $4.3 \times 10^{-14}$  at SF2. The two further realizations behave alike: with the correction the cumulative change is  $10^{-14}$  to  $10^{-16}$  while the width remains 21.2–21.3 cells (SF4) and 23.8–26.3 cells (SF2) above the reference, with position errors below one cell (Table 6). The corresponding transition widths nevertheless remain 18.36 and 25.65 cells above the reference. Global

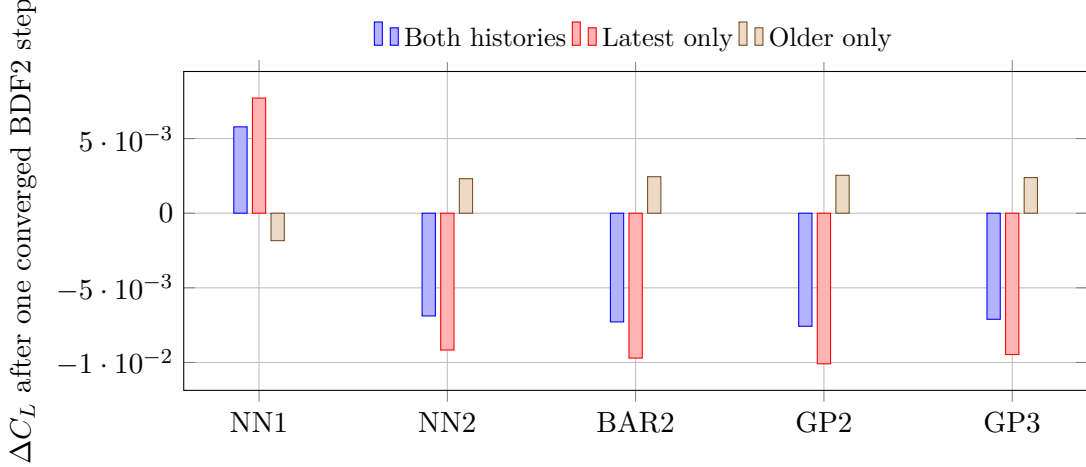


Figure 2: First-step lift response for the controlled Euler NACA 0012 BDF2 restart. The same target-referenced spatial discrepancy is inserted into both histories, only the latest history, or only the older history. The nonzero older-only response demonstrates that the latest state alone is not a sufficient restart diagnostic.

conservation therefore removes the scalar integral defect without reconstructing the lost spatial distribution. This result does not imply that Galerkin projection is uniquely capable of preserving the profile: NN2, NN3, BAR3, and GP3 also remain sharply resolved in the 400-transfer SF4 comparison. It does show that a zero global integral defect is insufficient to diagnose repeated-remapping fidelity.

The same hierarchy is visible in the interpolation-only accumulation. At 80 transfers NN1 changes the density integral by +0.0105%, compared with +0.0020% for BAR2 and +0.0003% for GP2. At the more frequent SF4 setting the NN1 change reaches +0.0500%, while GP2 remains at −0.0003%. These signed totals can contain cancellation and are not used as drift rates; their role is to quantify the imbalance introduced at the transfer events themselves.

#### 5.4 A BDF2 restart depends on both transferred histories

The steady Euler NACA 0012 experiment isolates the temporal consequence of a mesh transfer without introducing a physical unsteadiness. The target-referenced state discrepancy after transfer is approximately  $7.0 \times 10^{-4}$  for every second- and third-order method, with less than 3% spread, whereas NN1 gives  $1.47 \times 10^{-3}$ . This clustering is important: it prevents the subsequent response from being interpreted as a simple ranking of interpolation accuracy, because the target-referenced discrepancy also contains the native difference between the two discrete steady solutions.

The controlled history experiment nevertheless gives a direct result about BDF2. Figure 2 shows the first converged lift response when the same spatial discrepancy is placed in both histories, only the latest history, or only the older history. The older-only configuration has  $\epsilon^n = 0$ , yet produces a finite response for every transfer. For GP2, for example,  $\Delta C_L = +2.539 \times 10^{-3}$  in the older-only case, compared with  $-7.575 \times 10^{-3}$  for the full-history case. A metric based only on the latest transferred state would therefore predict zero forcing for a case that demonstrably changes the next physical step.

The measured response ratios are consistent with the coefficients in  $\eta^n = 2\epsilon^n - \epsilon^{n-1}/2$ . Across the seven tested transfers, the latest/full state-norm ratio is 1.3330–1.3333 and the older/full magnitude is 0.3335–0.3341. State-vector superposition defects are 1.8–3.5%; lift superposition defects are

0.28–1.49%. The test therefore supports the directional linearization used in Sec. 3, but does not validate a general target resolvent for arbitrary perturbation directions.

The common-trial residual check also exposes an implementation detail that is hidden by a scalar norm. NN and BAR variants recover the predicted history forcing to approximately  $10^{-7}$  relative error. For GP2 and GP3, the larger whole-vector discrepancy is localized to the 370 slip-wall vertices and to wall-normal momentum. The measured normal mismatch equals the projected normal forcing ( $2.125 \times 10^{-6}$  for GP2 and  $6.066 \times 10^{-7}$  for GP3), while the tangential mismatch is only  $O(10^{-11})$ . The receiving solver removes wall-normal momentum through the slip condition; after this boundary admissibility operation, the tangential/interior identity is again satisfied at approximately  $10^{-7}$ . Thus the algebraic BDF2 identity remains the correct starting point, but the physically admissible restart pipeline includes boundary-condition projection as well as interpolation.

Two further restarts test the algebra directly (Tables 7 and 8). With the histories  $\epsilon^n = e/4$  and  $\epsilon^{n-1} = e$ , for which  $\eta^n = 2\epsilon^n - \epsilon^{n-1}/2$  vanishes although both histories differ from the native state, the first converged step reproduces the native continuation: the lift differs by  $7.7 \times 10^{-9}$  (GP2) and  $2.1 \times 10^{-9}$  (NN1), and the state by  $2 \times 10^{-4}$  of  $\|e\|$ , the iterative floor. The residual probe of that pair differs from the native probe by 0.750 of the forcing that the older history alone would produce, exactly the factor 3/4 implied by the trial state, which SU2 initializes with the perturbed latest level; the identity therefore holds once the trial states are matched. Scaling the same discrepancy by 1/2 and 1/4 for BAR2 and GP2 reduces the Euclidean superposition remainder of the state by a factor of 3.2 per halving, close to the quadratic remainder of a smooth nonlinearity, from  $1.8 \times 10^{-3}$  to  $1.6 \times 10^{-4}$ ; the lift remainder falls from  $2.2 \times 10^{-5}$  to  $1.8 \times 10^{-6}$  and then meets the force floor, and the response ratios approach 4/3 and 1/3 (1.3334 and 0.3333 at the smallest amplitude). In the volume-weighted norm the remainder stays at  $1.5\text{--}2.4 \times 10^{-5}$  for every amplitude because that norm is dominated by far-field cells at the iterative floor. The 1.8–3.5% superposition defect at full amplitude is thus the genuine nonlinear term of the local response.

Applying GC to the steady NACA transfer changes neither this conclusion nor the aerodynamic response materially. For NN1 the full-history lift response changes from  $5.7873 \times 10^{-3}$  to  $5.7924 \times 10^{-3}$ ; for BAR2 it changes from  $-7.2852 \times 10^{-3}$  to  $-7.2850 \times 10^{-3}$ . In these cases, enforcing the source integral alters a scalar component of the transferred field while leaving the force-relevant spatial discrepancy essentially intact.

## 5.5 Later aerodynamic response depends on the receiving flow problem

Two Euler experiments show that the first-step response is not the complete aerodynamic consequence of a transfer. Over the 400-step Euler window (approximately 0.047 chord flow-through times), GP2 has reproducible positive drag trends of  $6.84 \times 10^{-6}$  and  $6.38 \times 10^{-6}$  per 100 physical steps in the two realizations, whereas NN2 gives  $3.95 \times 10^{-5}$  and  $3.84 \times 10^{-5}$ . The two-seed mean ratio is about 5.8. This is a resolved difference for that Euler configuration, not a general ranking: lift trends are much less repeatable and the RANS calculations below do not reproduce the same method separation.

A single spatially localized GP2 transfer provides an independent test of delayed propagation. Increasing the launch distance from  $0.05c$  to  $0.20c$  delays the lift response, and pressure-shell measurements on the long  $0.20c$  case give threshold-dependent propagation speeds between 1.09 and 1.27 for the 10–75% crossings after excluding the immediate transfer footprint. These values bracket the nondimensional acoustic speed  $a_\infty = 1.183$ . The spread is sensitivity to the chosen detector, not statistical uncertainty. The result supports a dominant acoustic-scale propagation in the Euler case and explains why a force response can peak after the transfer rather than at the first converged step. In particular, NN1 reaches a lift discrepancy 4.66 times its first-step value five steps

Table 3: Maximum absolute source-relative integral defect during the repeated-remeshing RANS runs. Values compare source and target states at a transfer event; the last column is the ratio between the NN2 and GP2 maxima.

| Closure | Inventory    | NN2                   | GP2                   | NN2/GP2 |
|---------|--------------|-----------------------|-----------------------|---------|
| SA      | rho          | $8.41 \cdot 10^{-11}$ | $4.57 \cdot 10^{-13}$ | 184     |
| SA      | rho u        | $3.78 \cdot 10^{-9}$  | $5.76 \cdot 10^{-11}$ | 66      |
| SA      | rho v        | $1.65 \cdot 10^{-8}$  | $8.46 \cdot 10^{-11}$ | 195     |
| SA      | rho E        | $9.54 \cdot 10^{-11}$ | $5.83 \cdot 10^{-13}$ | 164     |
| SA      | rho nu-tilde | $1.7 \cdot 10^{-6}$   | $6.88 \cdot 10^{-9}$  | 247     |
| SST     | rho          | $8.25 \cdot 10^{-11}$ | $4.68 \cdot 10^{-13}$ | 177     |
| SST     | rho u        | $3.65 \cdot 10^{-9}$  | $2.89 \cdot 10^{-11}$ | 126     |
| SST     | rho v        | $1.64 \cdot 10^{-8}$  | $7.78 \cdot 10^{-11}$ | 211     |
| SST     | rho E        | $9.37 \cdot 10^{-11}$ | $6.75 \cdot 10^{-13}$ | 139     |
| SST     | rho k        | $2.22 \cdot 10^{-6}$  | $6 \cdot 10^{-8}$     | 37      |
| SST     | rho omega    | $7.98 \cdot 10^{-8}$  | $2.15 \cdot 10^{-9}$  | 37      |

after restart, whereas the second- and third-order cases have their largest recorded response at the first sampled step and then relax.

## 5.6 Fixed-body RANS: integral preservation improves, but the load ranking is not resolved

The RANS extension tests the transfer pipeline when turbulence histories and viscous stresses are part of the receiving problem. The executed-run startup tables establish a non-dimensional physical step of  $\Delta t^* = 0.347224$  for the configured  $10^{-3}$  s step; with  $U_\infty^* = 0.1$ , 400 steps span 13.9 chord flow-through times. The no-remeshing references remain effectively stationary on this interval, so the case-minus-reference histories isolate the response associated with the prescribed mesh replacements and transferred states on the chosen discretization.

The transfer-integral comparison is clear. Table 3 reports the maximum instantaneous source-to-target defect across the repeated-remeshing histories. GP2 reduces the monitored mean-flow integral defects by roughly two orders of magnitude relative to NN2. The maximum defect in the transported turbulence inventories is reduced by a factor of 247 for  $\rho\tilde{\nu}$  in SA and by factors of about 37 for both  $\rho k$  and  $\rho\omega$  in SST. These are transfer-event diagnostics; for SST the archived integral is evaluated before the primitive-variable positivity floor is applied. Turbulence production, destruction, and transport subsequently change the corresponding integrals and are not counted as remapping errors.

The aerodynamic histories do not show an analogous separation. Across the eight method/seed/closure combinations, the RMS total-drag deviations from the corresponding no-remeshing reference are  $(6.07\text{--}6.79) \times 10^{-5}$ . The paired NN2–GP2 difference has RMS  $(0.77\text{--}1.11) \times 10^{-5}$ , and the ordering of the two methods changes with closure and deformation realization. These statistics show that the two returned solutions are distinguishable, but not that either transfer is aerodynamically more accurate.

The numerical controls explain why a stronger ranking is not justified. In the production runs, 144–171 of 398 SA steps and 277–333 of 398 SST steps reach the 800-iteration dual-time cap. For SA, raising the cap to 1600 brings every step of the control window to the target residual and changes the case-minus-reference lift and drag by 0.2–0.3% of their RMS and the paired NN2–GP2 difference by 8–11%: the SA comparison is converged with respect to the inner loop well below the paired difference. For SST the inner loop has a floor: with caps of 1600 and 6400, 28–30 of 38 steps still

stop at the cap, and the terminal density and  $\omega$  residuals stay at  $10^{-11.1}$ – $10^{-11.4}$  and  $10^{-6.8}$ – $10^{-7.0}$  regardless of the cap. The SST case-minus-reference responses move by 4–9% of their RMS at each doubling of the cap without decreasing, and the paired difference moves by 78% of its own RMS between the caps of 1600 and 6400. The SST paired load difference is therefore undetermined at the level of the difference itself.

Temporal sensitivity is larger still. Halving the physical step while replaying the same three mesh events at the same physical times leaves the first-step response after each event unchanged to three digits, but changes the individual lift response by 41–42% of its production RMS for SA and by 100% for SST, and the drag response by 24% and 40%; the time-aligned NN2–GP2 difference changes by 1.4–1.5 times its production RMS for SA and by 1.5–1.6 times for SST, while its magnitude stays of the same order. Quartering the SA step does not bring the sequence into an asymptotic range: the change from  $\Delta t/2$  to  $\Delta t/4$  equals the change from  $\Delta t$  to  $\Delta t/2$  for the individual responses (ratios 0.85–1.04) and is twice as large for the paired difference, whose RMS grows from  $1.2 \times 10^{-5}$  to  $3.5 \times 10^{-5}$ . The detailed paired transient is therefore not time-step converged within the tested range for either closure.

A single finer no-transfer mesh provides a separate spatial-sensitivity check. Relative to the production mesh, the finer mesh changes  $C_L$  by  $-0.80\%$  for SA and  $-1.04\%$  for SST and changes  $C_D$  by  $-1.77\%$  and  $-1.55\%$ , respectively. Because only one refinement is available, these differences are not a discretization-uncertainty estimate or a grid-convergence result. They do show that the absolute RANS loads remain mesh-sensitive at a level larger than the remeshing-induced method difference. This does not invalidate the paired NN2–GP2 comparison on a common mesh; it prevents the present data from supporting an absolute aerodynamic-accuracy claim. Repeating the SA control window on the finer mesh with the same seed and motion parameters, that is at equal remeshing severity on different nodes, reduces the remeshing response itself (lift RMS  $2.6 \times 10^{-4}$  to  $1.1 \times 10^{-4}$ , drag  $6.6 \times 10^{-5}$  to  $5.4 \times 10^{-5}$ ) while the paired difference stays at 3–8% of the response on both meshes; no ranking emerges with refinement.

The defensible RANS conclusion is therefore deliberately limited. GP2 returns states with substantially smaller monitored integral defects, whereas the available load calculations do not resolve a corresponding aerodynamic advantage over NN2. The application demonstrates that a transfer property measured before restart need not map monotonically to an aerodynamic output after the receiving solver, and it identifies temporal resolution in SA and the  $\omega$ -equation floor in SST as the controls that leave the small paired load difference unresolved.

## 6 Discussion and conclusions

The results support a narrower and more useful conclusion than a universal ranking of interpolation schemes. A remeshing event introduces several distinct numerical questions: how accurately the source field is represented on the new mesh; whether bounds, admissibility, and monitored integrals are preserved; whether every history required by the time integrator is transferred consistently; and how the receiving discrete operator converts the resulting discrepancy into state and output errors. No single scalar diagnostic orders all of these effects.

First, global conservation is insufficient as a spatial-quality criterion. In the repeated Riemann test, increasing transfer frequency or deformation amplitude primarily broadens the density transition rather than translating it. NN1 produces 21–27 cells of excess width across the high-frequency realizations and BAR2 ranges from about 3 cells at SF4 to 23 cells at SF2, while GP2 remains below 0.06 cell. The same hierarchy is recovered for three independent deformation sequences. Correcting NN1 reduces the cumulative density-integral defect to roundoff but leaves 18–26 cells of excess

width. A global integral constraint therefore fixes the total amount of a conserved quantity without determining its spatial distribution. The success of higher-order NN/BAR variants in the same experiment also rules out attributing the result uniquely to “local conservation.”

Second, a BDF2 restart is intrinsically a two-history problem. The older-only NACA configuration produces a finite state and force response even though the latest target history is exact, and the latest/full and older/full response ratios recover the BDF2 coefficients to within a few tenths of a percent in norm. The remaining 1.8–3.5% vector superposition defects delimit the accuracy of the local linear interpretation. The residual audit further shows that boundary enforcement belongs to the effective transfer operator: Galerkin projection introduces a wall-normal momentum history component at the slip wall, and the receiving boundary condition removes it. Limiting, positivity enforcement, boundary projection, and representation changes must therefore be included when evaluating the state actually supplied to the time integrator.

Third, aerodynamic consequences depend on the receiving problem. In the repeated Euler airfoil calculation GP2 reduces the reproducible drag trend by approximately a factor of 5.8 relative to NN2, and the localized-transfer experiment shows a delayed response consistent with acoustic-scale propagation. The fixed-body RANS calculations do not reproduce an equally clear method separation. GP2 reduces the monitored turbulence-inventory defects by factors of roughly 37–247, yet NN2 and GP2 have similar aggregate drag deviations on the tested mesh sequences. For SA the paired lift/drag difference is converged with respect to the dual-time iteration cap but changes by more than its production RMS when the physical step is halved and again when it is quartered; for SST an  $\omega$ -residual floor on the moved grids leaves the paired difference undetermined at its own level. The finer viscous meshes shift the absolute loads by 1–2% and reduce the remeshing response, without separating the two transfers. The present RANS evidence therefore supports the transfer-state and integral comparison, but not a quantitative aerodynamic ranking of NN2 and GP2.

The practical implication is that interpolation order and conservation should be treated as necessary diagnostics rather than stand-alone selection criteria. A defensible remeshing workflow should verify smooth-field accuracy, boundedness and admissibility, source-to-target integral changes, compatibility of every multistep history, boundary processing, and the convergence properties of the receiving physical step. When the output difference between transfer methods is smaller than the temporal sensitivity of the restarted calculation, improving the interpolation alone cannot be shown to improve the final aerodynamic prediction.

Within the tested configurations, four conclusions are supported directly by the evidence. The implemented transfers recover their intended smooth-field orders. A bounded global correction can enforce the monitored solver integral without controlling cumulative remapping diffusion. Both BDF2 history levels contribute directly to the restart transient. Finally, better preservation of transferred integrals does not imply a universal improvement in aerodynamic loads. The Riemann, Euler-airfoil, and RANS cases play complementary roles in establishing these statements: the first isolates repeated spatial transport, the second isolates multistep-history forcing and delayed response, and the third demonstrates how transfer-state differences interact with turbulence transport, viscous forces, and finite solver resolution.

## A Supporting results

### A.1 Vertex restriction of a quadratic projection

The conservation property of the complete Galerkin field does not automatically transfer to a vertex-only restart representation. Consider  $f(x) = x^2$  on a uniform partition of  $[0, 1]$ . A quadratic

finite-element space represents  $f$  exactly, so the  $P^2$  projection itself has no approximation error. If only the vertex values are retained and integrated with the corresponding  $P^1$  nodal weights, however,

$$Q_h(f) = \frac{1}{3} + \frac{h^2}{6}, \quad (30)$$

while the exact integral is  $1/3$ . Thus an exact quadratic projected field can acquire a second-order lumped-integral discrepancy solely through representation change. This is why the complete Galerkin field, the vertex restart state, and any subsequent correction are treated as distinct objects in the main text.

## Riemann frequency and amplitude matrix

Table 4 gives the complete frequency comparison used to support Sec. 5.3. Position and width use one common extraction procedure within this dataset.

Table 4: Transition position and width changes for the 80- and 400-transfer Riemann runs at SF4.

| Transfer | Transfers | $\Delta x/h$ | $\Delta w/h$ | $\sum \Delta I_\rho$ |
|----------|-----------|--------------|--------------|----------------------|
| NN1      | 80        | 0.14         | 0.57         | $1.1 \cdot 10^{-4}$  |
| NN1      | 400       | 0.1          | 21.41        | $5 \cdot 10^{-4}$    |
| NN2      | 80        | -0.02        | -0.01        | $-8.8 \cdot 10^{-6}$ |
| NN2      | 400       | -0.04        | 0            | $-1.7 \cdot 10^{-5}$ |
| NN3      | 80        | 0.01         | -0.01        | $2.2 \cdot 10^{-5}$  |
| NN3      | 400       | 0.02         | 0            | $2.7 \cdot 10^{-5}$  |
| BAR2     | 80        | 0.01         | 0.25         | $2 \cdot 10^{-5}$    |
| BAR2     | 400       | -0.03        | 3.24         | $1.4 \cdot 10^{-4}$  |
| BAR3     | 80        | 0            | 0.01         | $-1.7 \cdot 10^{-5}$ |
| BAR3     | 400       | 0            | 0.07         | $-5.2 \cdot 10^{-6}$ |
| GP2      | 80        | 0.01         | 0            | $2.6 \cdot 10^{-6}$  |
| GP2      | 400       | 0.01         | 0            | $-3 \cdot 10^{-6}$   |
| GP3      | 80        | -0.01        | 0            | $-2.1 \cdot 10^{-5}$ |
| GP3      | 400       | -0.01        | 0.02         | $1.8 \cdot 10^{-5}$  |

## NACA history-response ratios

Table 5: Directional BDF2 response ratios and superposition defects. The ideal linear-history values are  $4/3$  for latest/full and  $1/3$  in magnitude for older/full.

| Transfer | $e_{sup}^U$          | $\ \delta_n\ /\ \delta_f\ $ | $\ \delta_{n-1}\ /\ \delta_f\ $ | $e_{sup}^{C_L}$      | $\Delta C_{L,n}/\Delta C_{L,f}$ | $\Delta C_{L,n-1}/\Delta C_{L,f}$ |
|----------|----------------------|-----------------------------|---------------------------------|----------------------|---------------------------------|-----------------------------------|
| NN1      | $1.81 \cdot 10^{-2}$ | 1.3333                      | 0.3335                          | $1.49 \cdot 10^{-2}$ | 1.3329                          | -0.3179                           |
| NN2      | $3.53 \cdot 10^{-2}$ | 1.3331                      | 0.3341                          | $4.2 \cdot 10^{-3}$  | 1.3318                          | -0.336                            |
| NN3      | $3.4 \cdot 10^{-2}$  | 1.3331                      | 0.334                           | $5.29 \cdot 10^{-3}$ | 1.3309                          | -0.3362                           |
| BAR2     | $3.36 \cdot 10^{-2}$ | 1.3332                      | 0.334                           | $3.05 \cdot 10^{-3}$ | 1.3328                          | -0.3358                           |
| BAR3     | $3.34 \cdot 10^{-2}$ | 1.3332                      | 0.334                           | $2.83 \cdot 10^{-3}$ | 1.3329                          | -0.3357                           |
| GP2      | $3.39 \cdot 10^{-2}$ | 1.333                       | 0.334                           | $2.98 \cdot 10^{-3}$ | 1.3322                          | -0.3352                           |
| GP3      | $3.42 \cdot 10^{-2}$ | 1.3331                      | 0.334                           | $3.26 \cdot 10^{-3}$ | 1.3326                          | -0.3359                           |

## Riemann deformation realizations

Table 6 repeats the high-frequency stress comparison with two additional random deformation sequences at both amplitudes.

Table 6: Transition position error and width change after the high-frequency Riemann remeshing sequence for three independent deformation realizations. Values are expressed in baseline cells; SF2 permits twice the displacement of SF4.

| Transfer | GC     | SF | Seed | $\Delta x/h$ | $\Delta w/h$ | $\sum \Delta I_\rho$  |
|----------|--------|----|------|--------------|--------------|-----------------------|
| BAR2     | off    | 2  | 1    | -0.06        | 22.91        | $2.7 \cdot 10^{-4}$   |
| BAR2     | off    | 2  | 2    | -0.03        | 22.98        | $3.1 \cdot 10^{-4}$   |
| BAR2     | off    | 2  | 3    | -0.07        | 22.9         | $1.9 \cdot 10^{-4}$   |
| BAR2     | off    | 4  | 1    | -0.03        | 3.24         | $1.4 \cdot 10^{-4}$   |
| BAR2     | off    | 4  | 2    | 0            | 3.2          | $1.7 \cdot 10^{-4}$   |
| BAR2     | off    | 4  | 3    | -0.04        | 3.12         | $6.6 \cdot 10^{-5}$   |
| GP2      | off    | 2  | 1    | 0.17         | 0.06         | $-1.1 \cdot 10^{-5}$  |
| GP2      | off    | 2  | 2    | 0.19         | 0.06         | $-1 \cdot 10^{-5}$    |
| GP2      | off    | 2  | 3    | 0.19         | 0.03         | $-9.7 \cdot 10^{-6}$  |
| GP2      | off    | 4  | 1    | 0.01         | 0            | $-3 \cdot 10^{-6}$    |
| GP2      | off    | 4  | 2    | 0.02         | 0            | $-3.1 \cdot 10^{-6}$  |
| GP2      | off    | 4  | 3    | 0.01         | 0.01         | $-2.6 \cdot 10^{-6}$  |
| NN1      | Global | 2  | 1    | -0.91        | 25.65        | $4.3 \cdot 10^{-14}$  |
| NN1      | Global | 2  | 2    | -0.66        | 23.8         | $2.7 \cdot 10^{-14}$  |
| NN1      | Global | 2  | 3    | 0            | 26.32        | $1.2 \cdot 10^{-14}$  |
| NN1      | Global | 4  | 1    | -0.15        | 18.36        | $5.8 \cdot 10^{-14}$  |
| NN1      | Global | 4  | 2    | -0.01        | 21.28        | $-9.3 \cdot 10^{-16}$ |
| NN1      | Global | 4  | 3    | -0.12        | 21.18        | $9.4 \cdot 10^{-14}$  |
| NN1      | off    | 2  | 1    | -0.38        | 26.36        | $1.3 \cdot 10^{-3}$   |
| NN1      | off    | 2  | 2    | -0.29        | 24.98        | $-1.6 \cdot 10^{-3}$  |
| NN1      | off    | 2  | 3    | 0.44         | 27.14        | $4.9 \cdot 10^{-4}$   |
| NN1      | off    | 4  | 1    | 0.1          | 21.41        | $5 \cdot 10^{-4}$     |
| NN1      | off    | 4  | 2    | 0.3          | 22.89        | $-8.9 \cdot 10^{-4}$  |
| NN1      | off    | 4  | 3    | 0.2          | 22.75        | $8.3 \cdot 10^{-4}$   |

## BDF2 cancellation and amplitude checks

Table 7: History-cancellation restart ( $\epsilon^n = e/4$ ,  $\epsilon^{n-1} = e$ , so  $\eta^n = 0$ ) against the native BDF2 continuation: first-step lift difference, largest lift difference over the 50-step window (the older level still carries  $e/4$  after the first step), step-2 state difference relative to  $\|e\|$ , and the initial-residual difference relative to the forcing of the older history alone (the trial-state offset predicts  $3/4$ ).

| Transfer | $\Delta C_L$ , step 2 | $\max  \Delta C_L $ , window | $\ \delta U\ /\ e\ $ | $\ r\ /\ p_{\text{older}}\ $ |
|----------|-----------------------|------------------------------|----------------------|------------------------------|
| GP2      | $7.7 \cdot 10^{-9}$   | $6.3 \cdot 10^{-4}$          | $2.4 \cdot 10^{-4}$  | 0.75                         |
| NN1      | $2.1 \cdot 10^{-9}$   | $3.1 \cdot 10^{-3}$          | $2.1 \cdot 10^{-4}$  | 0.75                         |

Table 8: Superposition remainder of the first-step response against the perturbation amplitude  $\lambda$  (discrepancy  $\lambda e$  in both, the latest, or the older history): Euclidean state remainder and its relative value, lift remainder, and the response ratios (ideal  $4/3$  and  $-1/3$ ).

| Transfer | $\lambda$ | $\ \delta_f - \delta_n - \delta_{n-1}\ $ | relative            | lift remainder      | $\Delta C_{L,n}/\Delta C_{L,f}$ | $\Delta C_{L,n-1}/\Delta C_{L,f}$ |
|----------|-----------|------------------------------------------|---------------------|---------------------|---------------------------------|-----------------------------------|
| BAR2     | 1         | $1.8 \cdot 10^{-3}$                      | $7.8 \cdot 10^{-3}$ | $2.2 \cdot 10^{-5}$ | 1.3328                          | -0.3358                           |
| BAR2     | 0.5       | $5.4 \cdot 10^{-4}$                      | $4.8 \cdot 10^{-3}$ | $4.7 \cdot 10^{-6}$ | 1.3328                          | -0.3341                           |
| BAR2     | 0.25      | $1.6 \cdot 10^{-4}$                      | $2.8 \cdot 10^{-3}$ | $1.8 \cdot 10^{-6}$ | 1.3329                          | -0.3339                           |
| GP2      | 1         | $1.9 \cdot 10^{-3}$                      | $8.3 \cdot 10^{-3}$ | $2.3 \cdot 10^{-5}$ | 1.3322                          | -0.3352                           |
| GP2      | 0.5       | $5.7 \cdot 10^{-4}$                      | $5.1 \cdot 10^{-3}$ | $3.6 \cdot 10^{-6}$ | 1.333                           | -0.334                            |
| GP2      | 0.25      | $1.6 \cdot 10^{-4}$                      | $2.9 \cdot 10^{-3}$ | $1.5 \cdot 10^{-6}$ | 1.3329                          | -0.3337                           |

## RANS numerical-control diagnostics

The runtime logs give  $\Delta t^* = 0.347224$  and  $U_\infty^* = 0.1$ . Table 9 reports how often the repeated-remeshing calculations reach the 800-iteration dual-time cap.

Table 9: Inner-iteration behavior of the repeated-remeshing RANS calculations. “At cap” counts physical steps that reach the maximum inner-iteration count before satisfying the configured density-residual criterion.

| Closure | Transfer | Seed | Steps | At cap | median $\log_{10} \text{rms}(\rho)$ |
|---------|----------|------|-------|--------|-------------------------------------|
| SA      | GP2      | 1    | 398   | 144    | -12                                 |
| SA      | GP2      | 2    | 398   | 169    | -12                                 |
| SA      | NN2      | 1    | 398   | 156    | -12                                 |
| SA      | NN2      | 2    | 398   | 171    | -12                                 |
| SST     | GP2      | 1    | 398   | 333    | -11.1                               |
| SST     | GP2      | 2    | 398   | 284    | -11.31                              |
| SST     | NN2      | 1    | 398   | 323    | -11.24                              |
| SST     | NN2      | 2    | 398   | 277    | -11.24                              |

Table 10 summarizes the two solver controls over their common physical window.  $D_m$  denotes the case-minus-reference response for transfer method  $m$ , and  $P = \text{NN2} - \text{GP2}$  denotes the paired method difference. The cap controls (1600 for both closures, 6400 for SST), the half-step control (both closures), the quarter-step control (SA) and the fine-mesh window (SA) are listed.

Table 10: Sensitivity of the repeated-remeshing RANS response to the dual-time iteration cap (800 to 1600 and 6400), to the physical step (halved and quartered, with the same remeshing events replayed at the same physical times), and to the mesh (the same window on the finer mesh, own random field). The last column is the RMS change relative to the production RMS of the same quantity.

| Closure | Control     | Quantity | Coeff. | RMS, production      | RMS change           | Change / production |
|---------|-------------|----------|--------|----------------------|----------------------|---------------------|
| SA      | cap1600     | GP2      | CL     | $2.63 \cdot 10^{-4}$ | $6.53 \cdot 10^{-7}$ | 0.002               |
| SA      | cap1600     | GP2      | CD     | $6.56 \cdot 10^{-5}$ | $1.33 \cdot 10^{-7}$ | 0.002               |
| SA      | cap1600     | NN2      | CL     | $2.6 \cdot 10^{-4}$  | $8.79 \cdot 10^{-7}$ | 0.003               |
| SA      | cap1600     | NN2      | CD     | $6.61 \cdot 10^{-5}$ | $6.69 \cdot 10^{-8}$ | 0.001               |
| SA      | cap1600     | P        | CL     | $1.21 \cdot 10^{-5}$ | $1.32 \cdot 10^{-6}$ | 0.109               |
| SA      | cap1600     | P        | CD     | $2.19 \cdot 10^{-6}$ | $1.83 \cdot 10^{-7}$ | 0.084               |
| SA      | halfstep    | GP2      | CL     | $2.63 \cdot 10^{-4}$ | $1.11 \cdot 10^{-4}$ | 0.423               |
| SA      | halfstep    | GP2      | CD     | $6.56 \cdot 10^{-5}$ | $1.6 \cdot 10^{-5}$  | 0.244               |
| SA      | halfstep    | NN2      | CL     | $2.6 \cdot 10^{-4}$  | $1.06 \cdot 10^{-4}$ | 0.408               |
| SA      | halfstep    | NN2      | CD     | $6.61 \cdot 10^{-5}$ | $1.58 \cdot 10^{-5}$ | 0.24                |
| SA      | halfstep    | P        | CL     | $1.21 \cdot 10^{-5}$ | $1.67 \cdot 10^{-5}$ | 1.386               |
| SA      | halfstep    | P        | CD     | $2.19 \cdot 10^{-6}$ | $3.21 \cdot 10^{-6}$ | 1.468               |
| SA      | quarterstep | GP2      | CL     | $2.63 \cdot 10^{-4}$ | $1.63 \cdot 10^{-4}$ | 0.619               |
| SA      | quarterstep | GP2      | CD     | $6.56 \cdot 10^{-5}$ | $2.37 \cdot 10^{-5}$ | 0.362               |
| SA      | quarterstep | NN2      | CL     | $2.6 \cdot 10^{-4}$  | $1.53 \cdot 10^{-4}$ | 0.587               |
| SA      | quarterstep | NN2      | CD     | $6.61 \cdot 10^{-5}$ | $2.21 \cdot 10^{-5}$ | 0.334               |
| SA      | quarterstep | P        | CL     | $1.21 \cdot 10^{-5}$ | $3.76 \cdot 10^{-5}$ | 3.12                |
| SA      | quarterstep | P        | CD     | $2.19 \cdot 10^{-6}$ | $6.93 \cdot 10^{-6}$ | 3.169               |
| SA      | finemesh    | GP2      | CL     | $2.63 \cdot 10^{-4}$ | $1.97 \cdot 10^{-4}$ | 0.75                |
| SA      | finemesh    | GP2      | CD     | $6.56 \cdot 10^{-5}$ | $2.1 \cdot 10^{-5}$  | 0.321               |
| SA      | finemesh    | NN2      | CL     | $2.6 \cdot 10^{-4}$  | $1.99 \cdot 10^{-4}$ | 0.763               |
| SA      | finemesh    | NN2      | CD     | $6.61 \cdot 10^{-5}$ | $2.19 \cdot 10^{-5}$ | 0.331               |
| SA      | finemesh    | P        | CL     | $1.21 \cdot 10^{-5}$ | $1.43 \cdot 10^{-5}$ | 1.185               |
| SA      | finemesh    | P        | CD     | $2.19 \cdot 10^{-6}$ | $2.26 \cdot 10^{-6}$ | 1.033               |
| SST     | cap1600     | GP2      | CL     | $1.65 \cdot 10^{-4}$ | $6.17 \cdot 10^{-6}$ | 0.037               |
| SST     | cap1600     | GP2      | CD     | $5.45 \cdot 10^{-5}$ | $6.15 \cdot 10^{-7}$ | 0.011               |
| SST     | cap1600     | NN2      | CL     | $1.65 \cdot 10^{-4}$ | $6 \cdot 10^{-6}$    | 0.036               |
| SST     | cap1600     | NN2      | CD     | $5.51 \cdot 10^{-5}$ | $5.73 \cdot 10^{-7}$ | 0.01                |
| SST     | cap1600     | P        | CL     | $5.98 \cdot 10^{-6}$ | $8.19 \cdot 10^{-7}$ | 0.137               |
| SST     | cap1600     | P        | CD     | $1.59 \cdot 10^{-6}$ | $9.59 \cdot 10^{-8}$ | 0.06                |
| SST     | cap6400     | GP2      | CL     | $1.65 \cdot 10^{-4}$ | $1.15 \cdot 10^{-5}$ | 0.069               |
| SST     | cap6400     | GP2      | CD     | $5.45 \cdot 10^{-5}$ | $1.16 \cdot 10^{-6}$ | 0.021               |
| SST     | cap6400     | NN2      | CL     | $1.65 \cdot 10^{-4}$ | $1.42 \cdot 10^{-5}$ | 0.086               |
| SST     | cap6400     | NN2      | CD     | $5.51 \cdot 10^{-5}$ | $1.4 \cdot 10^{-6}$  | 0.025               |
| SST     | cap6400     | P        | CL     | $5.98 \cdot 10^{-6}$ | $4.07 \cdot 10^{-6}$ | 0.681               |
| SST     | cap6400     | P        | CD     | $1.59 \cdot 10^{-6}$ | $4.07 \cdot 10^{-7}$ | 0.257               |
| SST     | halfstep    | GP2      | CL     | $1.65 \cdot 10^{-4}$ | $1.65 \cdot 10^{-4}$ | 1                   |
| SST     | halfstep    | GP2      | CD     | $5.45 \cdot 10^{-5}$ | $2.16 \cdot 10^{-5}$ | 0.396               |
| SST     | halfstep    | NN2      | CL     | $1.65 \cdot 10^{-4}$ | $1.64 \cdot 10^{-4}$ | 0.994               |
| SST     | halfstep    | NN2      | CD     | $5.51 \cdot 10^{-5}$ | $2.08 \cdot 10^{-5}$ | 0.378               |
| SST     | halfstep    | P        | CL     | $5.98 \cdot 10^{-6}$ | $9.08 \cdot 10^{-6}$ | 1.52                |
| SST     | halfstep    | P        | CD     | $1.59 \cdot 10^{-6}$ | $2.54 \cdot 10^{-6}$ | 1.597               |

The finer-mesh comparison in Table 11 is a one-level sensitivity check, not a formal grid-convergence study. The two meshes use the same first off-wall spacing but differ in overall resolution.

Table 11: No-transfer steady loads on the production mesh (23,271 vertices) and a finer viscous mesh (32,102 vertices). The differences quantify the observed one-level mesh sensitivity; they are not a discretization-uncertainty estimate.

| Closure | Coefficient | Production mesh | Finer mesh | Difference            | %     |
|---------|-------------|-----------------|------------|-----------------------|-------|
| SA      | CL          | 0.676257        | 0.670838   | $-5.42 \cdot 10^{-3}$ | -0.8  |
| SA      | CD          | 0.010229        | 0.010048   | $-1.81 \cdot 10^{-4}$ | -1.77 |
| SA      | CDp         | 0.003559        | 0.003401   | $-1.58 \cdot 10^{-4}$ | -4.44 |
| SA      | CDv         | 0.006671        | 0.006647   | $-2.4 \cdot 10^{-5}$  | -0.36 |
| SST     | CL          | 0.675605        | 0.668596   | $-7.01 \cdot 10^{-3}$ | -1.04 |
| SST     | CD          | 0.009858        | 0.009705   | $-1.53 \cdot 10^{-4}$ | -1.55 |
| SST     | CDp         | 0.003469        | 0.003321   | $-1.48 \cdot 10^{-4}$ | -4.27 |
| SST     | CDv         | 0.006389        | 0.006385   | $-4 \cdot 10^{-6}$    | -0.06 |

## B Data provenance

All numerical values used in the Results and Conclusions are traceable to the versioned CSV files in `data/` and `RANS_PostProcessing/`; compact manuscript views in `generated/` are regenerated from those sources. Runtime normalization, mesh replay, mesh quality, code revisions, restart/checkpoint hashes, and inner-iteration controls are archived with the repository. The RANS control calculations use the same prescribed remeshing events at matched physical times where required. The repository release also records a clean-build and regeneration check so that the manuscript can be reproduced at the reduced-data level without relying on unversioned post-processing output.

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